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Nomadic Service Points

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Abstract

We consider the novel problem of dynamically assigning application sessions of mobile users or user groups to service points. Such assignments must balance the tradeoff between two conflicting goals. On the one hand, we would like to connect a user to the closest server, in order to reduce network costs and service latencies. On the other hand, we would like to minimize the number of costly session migrations, or handoffs, between service points. We tackle this problem using two approaches. First, we employ algorithmic online optimization to obtain algorithms whose worst-case performance is within a factor of the optimal. Next, we extend them with opportunistic versions that achieve excellent practical average performance and scalability. We conduct case studies of two settings where such algorithms are required: wireless mesh networks with mobile users, and wide-area groupware applications with or without mobility.

1 Introduction

Recent advances in network technology, along with the increasing demand for real-time networked applications, are bringing application service providers to deploy multiple geographically dispersed service points, or servers. This trend is expected to further expand with the explosion of new applications and the expansion of services to larger domains. In such settings, a given application session is typically associated with some server. In real-time applications, the association selection is driven by quality of service (QoS) considerations, which may depend, e.g., on the network distance of the user from the server. As many of these applications are becoming increasingly available to mobile users and dynamic user groups, the factors that dictate the server selection can vary with time. For example, due to a user's movement, a server providing optimal QoS at some point may later provide poor QoS, rendering it desirable to migrate the application session from one physical server to another. We therefore believe that many future distributed service infrastructures will employ *nomadic service points*, and will transparently manage such dynamic session assignments.

One important domain where nomadic service points can be exploited to serve mobile users is wireless mesh networks (WMNs) [1, 7, 9]. WMNs provide an increasingly popular solution for Internet access from residential areas with a limited wired infrastructure. These networks are built around multiple stationary wireless routers. Some of them, called access gateways, are wired to the Internet. A greedy routing protocol would route the traffic via the closest gateway as a mobile user